

```

# Task 1
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

pages = ["Home", "About", "Contact"]
visits = [120, 80, 50]

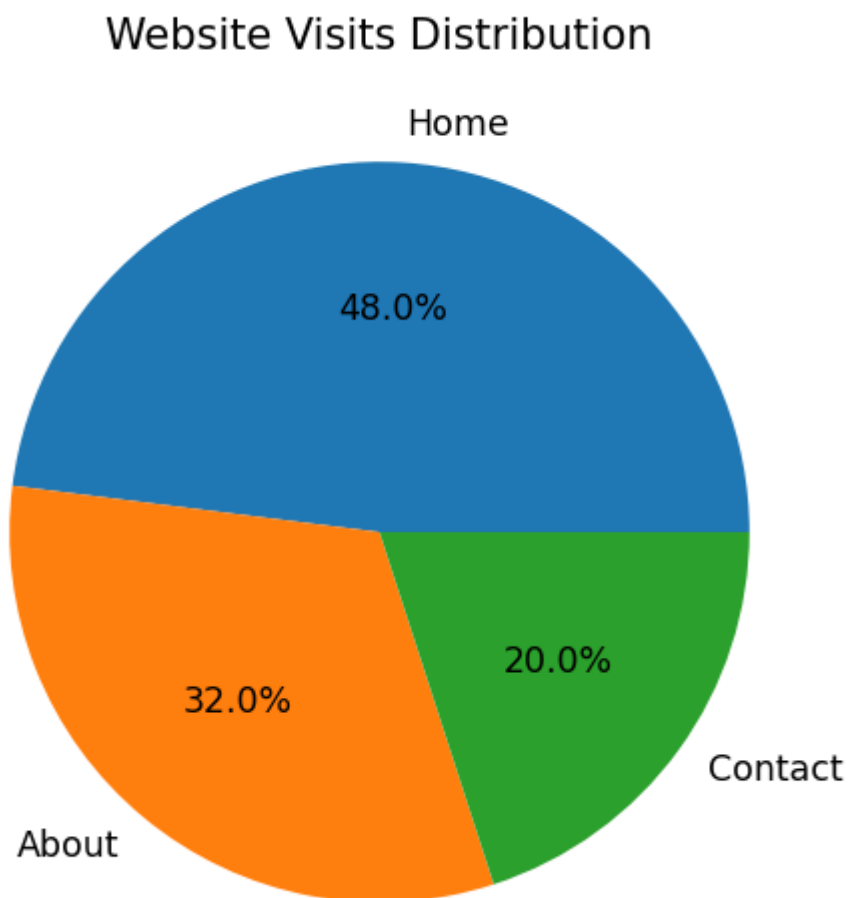
df = pd.DataFrame({"Page": pages, "Visits": visits})

total = np.sum(df["Visits"])
print("Total Visits:", total)

plt.pie(df["Visits"], labels=df["Page"], autopct="%1.1f%%")
plt.title("Website Visits Distribution")
plt.show()

===== RESTART:
Total Visits: 250

```



```
# Task 2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

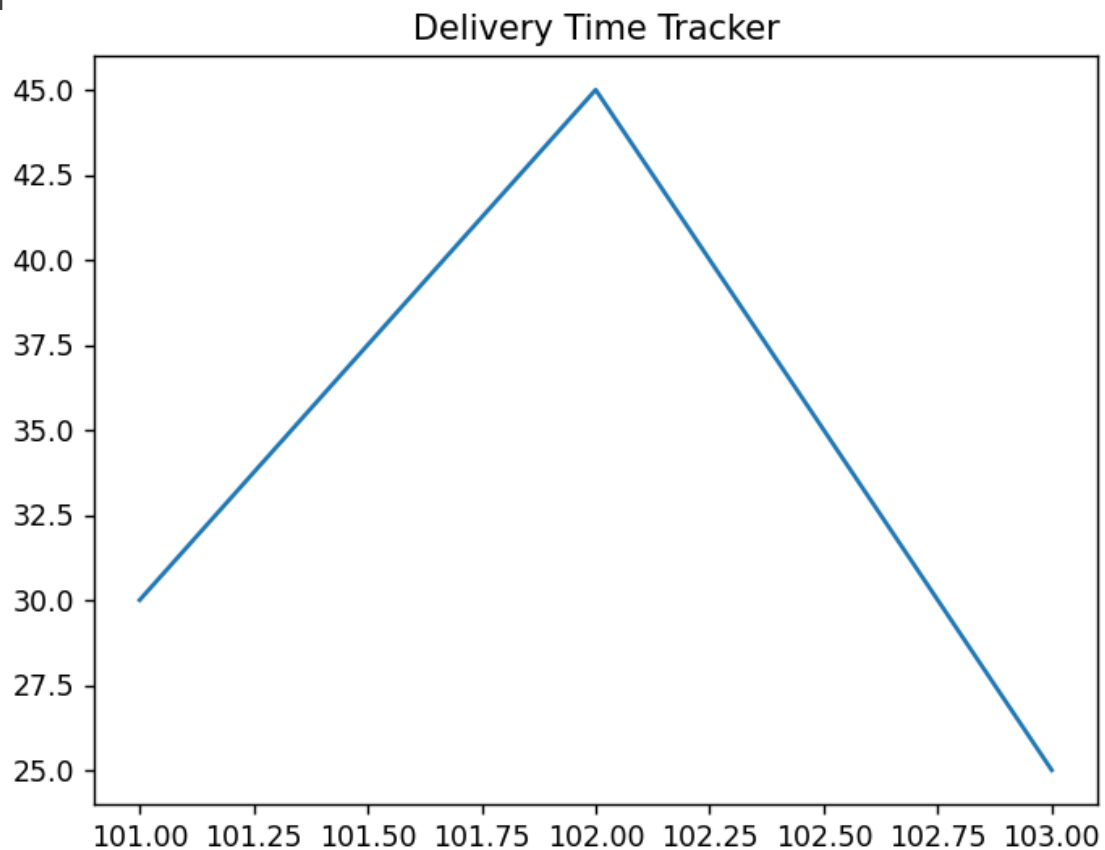
orders = [101, 102, 103]
delivery_time = [30, 45, 25]

df = pd.DataFrame({"OrderID": orders, "Time": delivery_time})

avg = np.mean(df["Time"])
print("Average Delivery Time:", avg)

plt.plot(df["OrderID"], df["Time"])
plt.title("Delivery Time Tracker")
plt.show()
```

Average Delivery Time: 33.333333333333336



```
# Task 3
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

employees = ["A", "B", "C"]
salary = [30000, 40000, 35000]

df = pd.DataFrame({"Employee": employees, "Salary": salary})

total = np.sum(df["Salary"])
avg = np.mean(df["Salary"])

print("Total Salary:", total)
print("Average Salary:", avg)

plt.bar(df["Employee"], df["Salary"])
plt.title("Employee Salary")
plt.show()

Total Salary: 105000
Average Salary: 35000.0
```



```
# Task 4
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

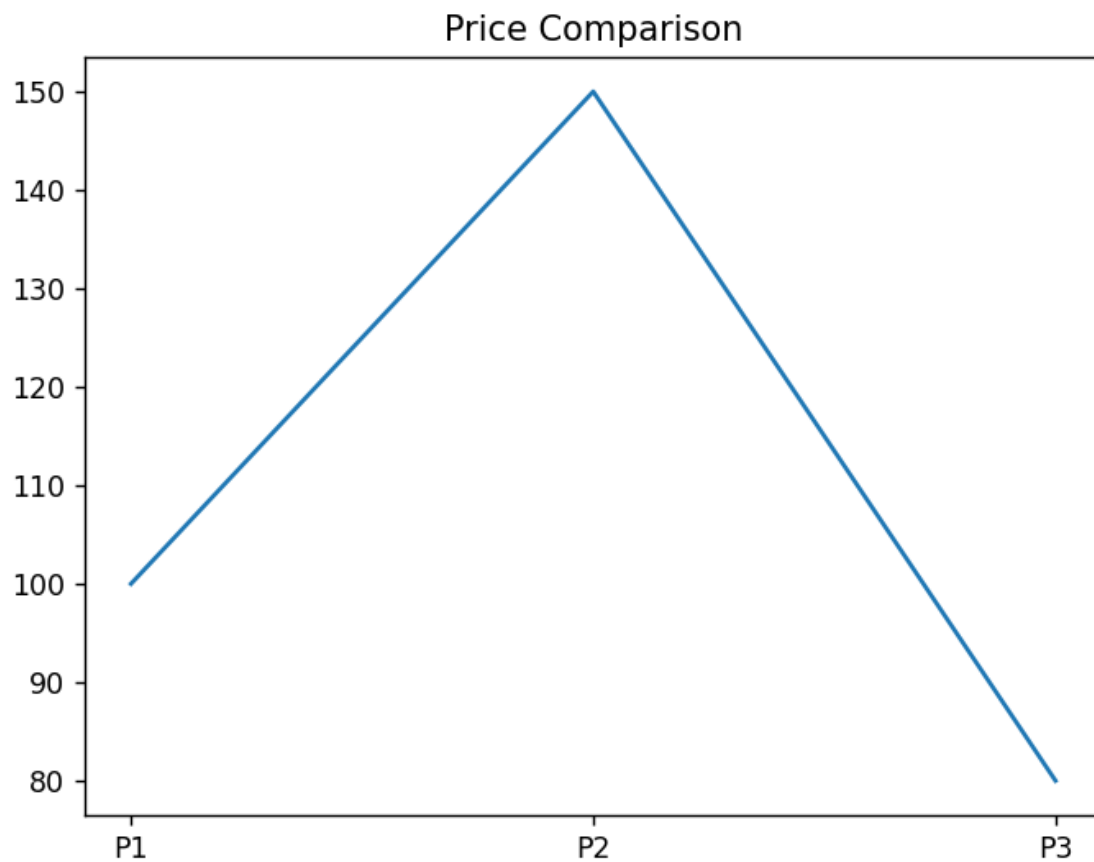
products = ["P1", "P2", "P3"]
price = [100, 150, 80]

df = pd.DataFrame({"Product": products, "Price": price})

print("Max Price:", np.max(df["Price"]))
print("Min Price:", np.min(df["Price"]))
print("Average Price:", np.mean(df["Price"]))

plt.plot(df["Product"], df["Price"])
plt.title("Price Comparison")
plt.show()
```

Max Price: 150
Min Price: 80
Average Price: 110.0



```
# Task 5
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

days = ["Mon", "Tue", "Wed", "Thu", "Fri"]
rainfall = [5, 10, 3, 8, 6]

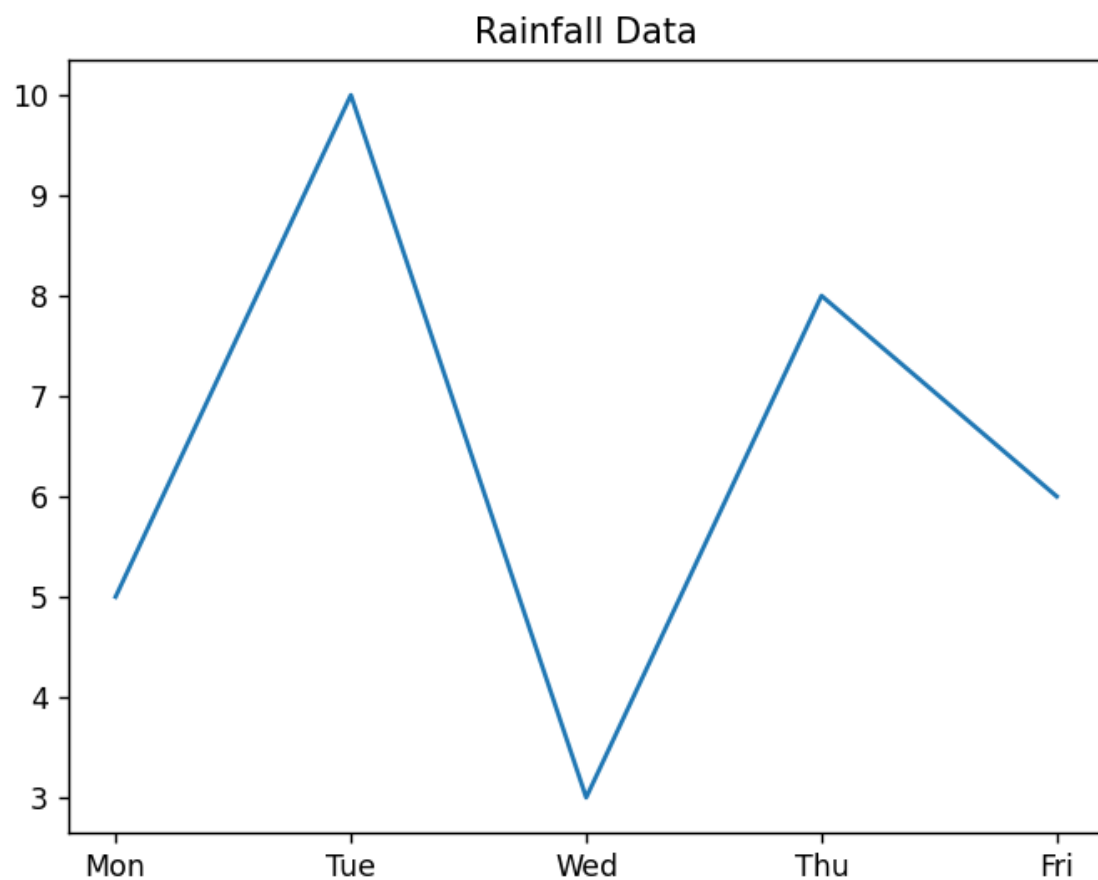
df = pd.DataFrame({"Day": days, "Rainfall": rainfall})

total = np.sum(df["Rainfall"])
avg = np.mean(df["Rainfall"])

print("Total Rainfall:", total)
print("Average Rainfall:", avg)

plt.plot(df["Day"], df["Rainfall"])
plt.title("Rainfall Data")
plt.show()

Total Rainfall: 32
Average Rainfall: 6.4
```



```
# Task 6
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

units = [10, 15, 12, 18, 20]

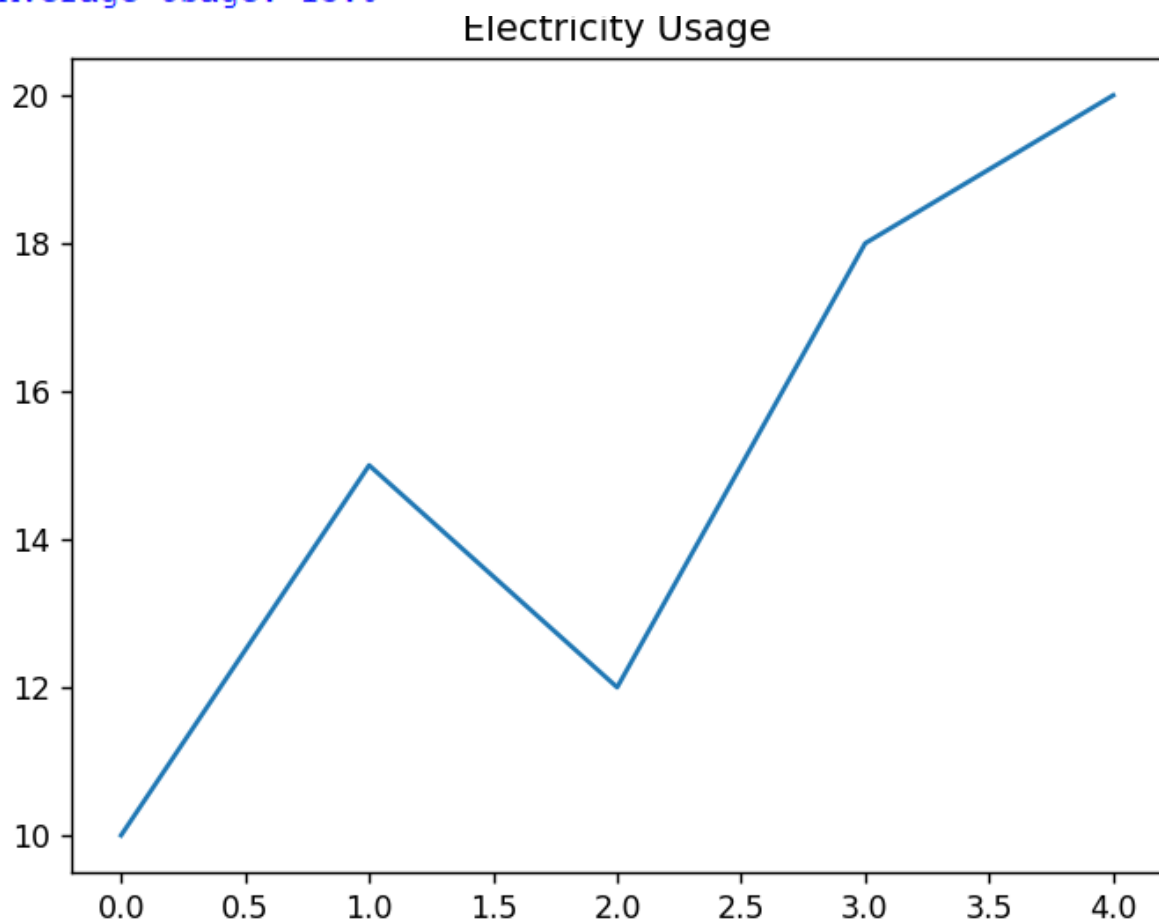
df = pd.DataFrame({"Units": units})

total = np.sum(df["Units"])
avg = np.mean(df["Units"])

print("Total Usage:", total)
print("Average Usage:", avg)

plt.plot(df["Units"])
plt.title("Electricity Usage")
plt.show()
```

Total Usage: 75
Average Usage: 15.0



```

# Task 7
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

apps = ["YouTube", "WhatsApp", "Instagram"]
time = [120, 60, 90]

df = pd.DataFrame({"App": apps, "Time": time})

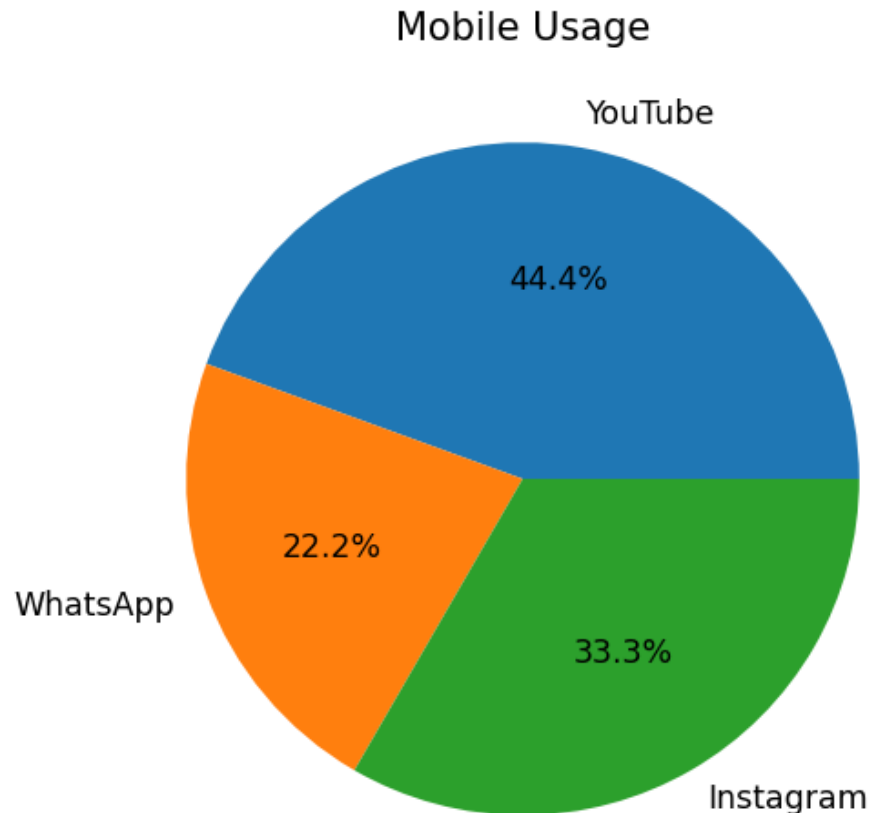
total = np.sum(df["Time"])
most_used = df.loc[df["Time"].idxmax()]

print("Total Screen Time:", total)
print("Most Used App:", most_used["App"])

plt.pie(df["Time"], labels=df["App"], autopct="%1.1f%%")
plt.title("Mobile Usage")
plt.show()

```

Total Screen Time: 270
Most Used App: YouTube



```

# Task 8
import pandas as pd
import numpy as np

TP = 70
TN = 50
FP = 20
FN = 10

data = [[TP, FP], [FN, TN]]
df = pd.DataFrame(data, columns=["Predicted Yes", "Predicted No"], index=["Actual Yes", "Actual No"])

total = TP + TN + FP + FN

accuracy = (TP + TN) / total
precision = TP / (TP + FP)
recall = TP / (TP + FN)
f1 = 2 * (precision * recall) / (precision + recall)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)

print("Confusion Matrix")
print(df)

```

```

Accuracy: 0.8
Precision: 0.7777777777777778
Recall: 0.875
F1 Score: 0.823529411764706
Confusion Matrix

```

	Predicted Yes	Predicted No
Actual Yes	70	20
Actual No	10	50